



ESMFold Finetuning on Toxin-Antitoxin Protein Sequences

CS 225:
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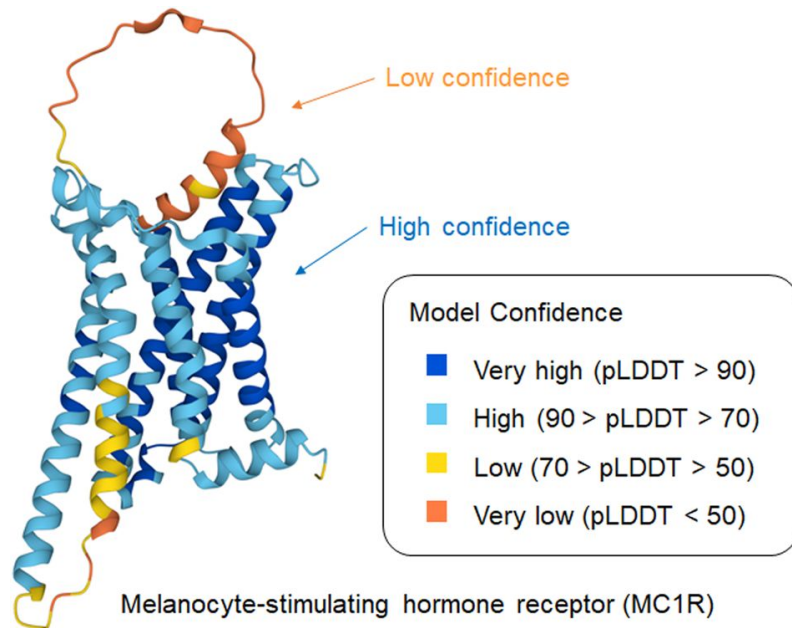


Motivation

- Protein structure prediction is important in bioinformatics since it allows us to see the shape of a protein structure and infer its functions
- **ESMFold** by Meta is a model that predicts protein structures from amino acid sequences
- Predictions are scored by pLDDT for local confidence of the structure
- Predicted structures with higher pLDDT scores as training data may improve model accuracy, as these represent more confident and structurally reliable predictions

Problem

Can we improve pLDDT scores on low-confidence protein predictions by fine-tuning ESMFold?





Data Used

- Toxin-Antitoxin Protein (Amino Acid) Sequences Dataset provided by Professor:
TASmania_hits_seqtk.faa

Two subsets were created to due to GPU memory limits in Google Colab for preprocessing of data

- Subset I (110/440 of original dataset)
 - Create subset of sequences by taking every 4th sequence in .faa file
 - Removed invalid amino acid sequences
 - Filtered sequences to be shorter than 128
- Subset II (105/440 of original dataset):
 - Created subset of sequences shorter than or equal to 128 amino acids long
 - Removed invalid amino acid sequences



Tools

- Google Colab (G4 High-Ram GPU) environment for training and inference with Python, PyTorch
- Meta ESMFold (Built on ESM-2) Model for generating PDB files, computing pLDDT scores, and fine-tuning
- Hugging Face Transformers for managing fine tuning and tokenizing data
- Biopython and Biotite for FASTA data parsing and computing pLDDT from PDB files



Methodology

1. Using ESMFold model, generated baseline PDB files and computed pLDDT confidence scores per PDB file
2. Filtered high-confidence proteins with $\text{pLDDT} \geq 0.7$ for fine-tuning, while saving low-confidence proteins with $\text{pLDDT} < 0.7$ for final evaluation
3. Froze most pretrained layers and then fine-tuned ESMFold, training on tokenized protein sequences that previously had high-confidence protein structures
 - a. Number of epochs = 10
4. Using the fine-tuned ESMFold model, regenerated PDB files on previously low-confidence protein structures and computed fine-tuned pLDDT scores to compare against baseline results

*Completed these steps for both data subsets



Demo of Colab

[https://colab.research.google.com/drive/1WZVAILGKN
Uv1P-bEhIZ-BwhUwFDVJ_cP?usp=sharing](https://colab.research.google.com/drive/1WZVAILGKNUv1P-bEhIZ-BwhUwFDVJ_cP?usp=sharing)



Results

Subset I (trained on 13 high-confidence sequences):

- Evaluated 10 low-confidence proteins with their baseline pLDDT < 0.7
- Average baseline pLDDT: 61.97
- Average fine-tuned pLDDT: 62.6, this is a 0.63 improvement
- 10 out of 10 proteins had improved pLDDT scores after fine-tuning

Subset II (trained on 60 high-confidence sequences):

- Evaluated 29 low-confidence proteins with their baseline pLDDT < 0.7
- Average baseline pLDDT: 56.22
- Average fine-tuned pLDDT: 56.2, this is a -0.02 decline
- 14 out of 29 proteins had improved pLDDT scores after fine-tuning

Results

Subset I:

[20/20 11:39, Epoch 10/10]		
Epoch	Training Loss	Validation Loss
1	0.000279	0.001334
2	0.000059	0.000633
3	0.000035	0.000391
4	0.000093	0.000334
5	0.000085	0.000363
6	0.000095	0.000387
7	0.000079	0.000455
8	0.000119	0.000504
9	0.000040	0.000558
10	0.000038	0.000563

Subset II:

[80/80 30:39, Epoch 10/10]		
Epoch	Training Loss	Validation Loss
1	0.000026	0.000498
2	0.000012	0.000657
3	0.000013	0.000599
4	0.000010	0.000625
5	0.000019	0.000674
6	0.000003	0.000593
7	0.000009	0.000613
8	0.000003	0.000627
9	0.000009	0.000578
10	0.000014	0.000577



Conclusion

- Lightweight fine-tuning of Meta's ESMFold model slightly improved the pLDDT confidence scores on low-confidence proteins for subset I, but not subset II
- This improvement was consistent across most evaluated proteins (10/10) for subset I, but only about half of the proteins for subset II (14/29)
- Both subset I and II are still ~25% of the original dataset
 - subset II included more sequences (~4.6x more) for training and (~3x more) for evaluation
- GPU memory limits in Google Collab required filtering protein sequences that were too long (over 128 amino acids) for training and evaluation
- In the future, with more less memory limits, we could use:
 - Larger Datasets
 - Longer Protein Sequences